Security on STM32 step by step

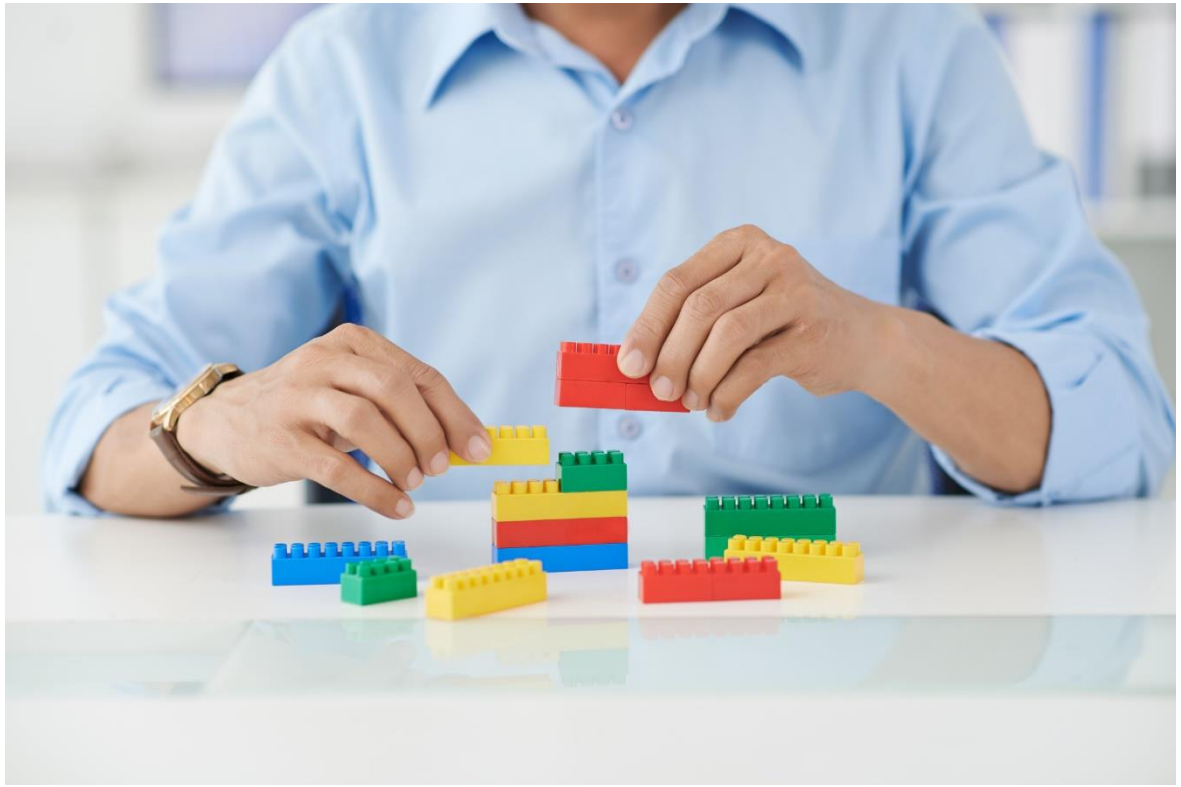


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Security Parts – how to

Assets

- All Security Parts are ordered in increasing difficulty level
- It is recommended to watch them in the proposed order
- Prerequisites and all the materials of each part are present within landing page (link visible within video description)



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Security Part1 – Introduction to security on STM32

Let’s try to find some answers to following questions:

- What is the security in embedded systems ?
- What we need to protect and why ?
- What is the online training offer at ST on this ?



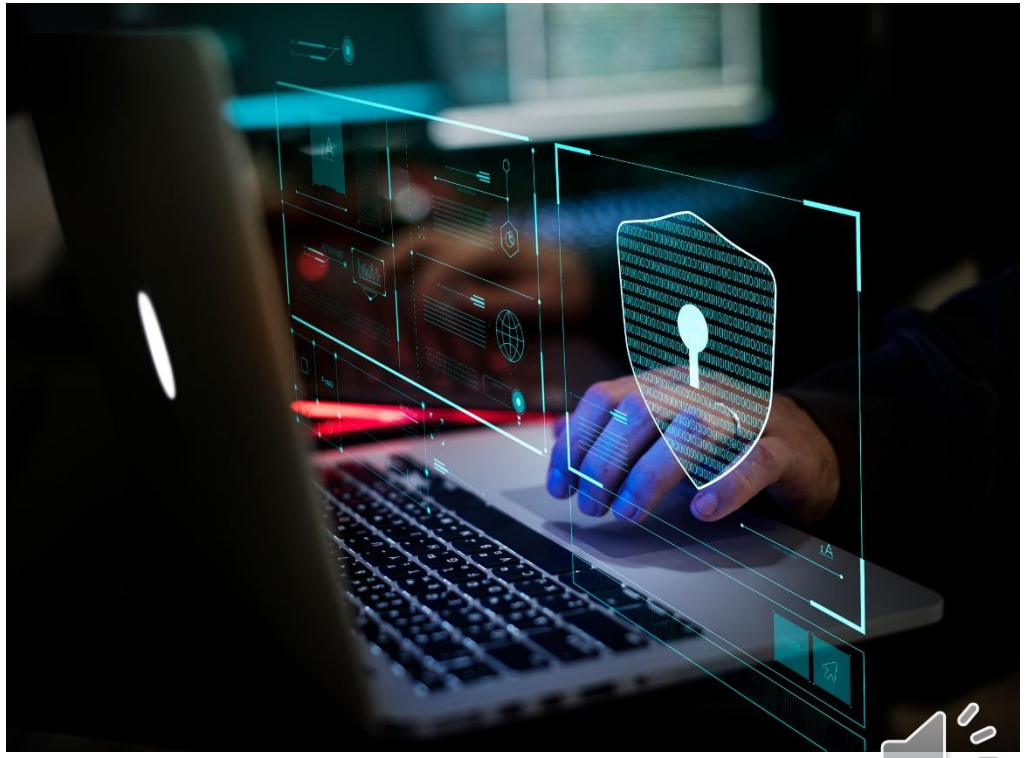
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Security Part2 – Basics of cryptography

Get basic knowledge and practice main cryptography concepts:

- Basics of symmetric cryptography
- Basics of assymetric cryptography
- Symmetric algorithms:
  - TDES, AES
- Assymetric algorithms:
  - RSA, ECC, DH, ECDH, IES
- Integrity and Authentication



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Security Part3 – STM32 security features

Get familiar with all security blocks available across STM32 lines and experience them when it’s possible

- Unique ID
- FLASH Memory protection:
  - Readout, Write, Proprietary Code
- Safe boot of the MCU
- Tamper detection
- Isolation items:
  - MPU, Firewall, TrustZone
- Support for cryptography:
  - TrueRNG, AES/CRYPT, PKA, HASH, Crypto library



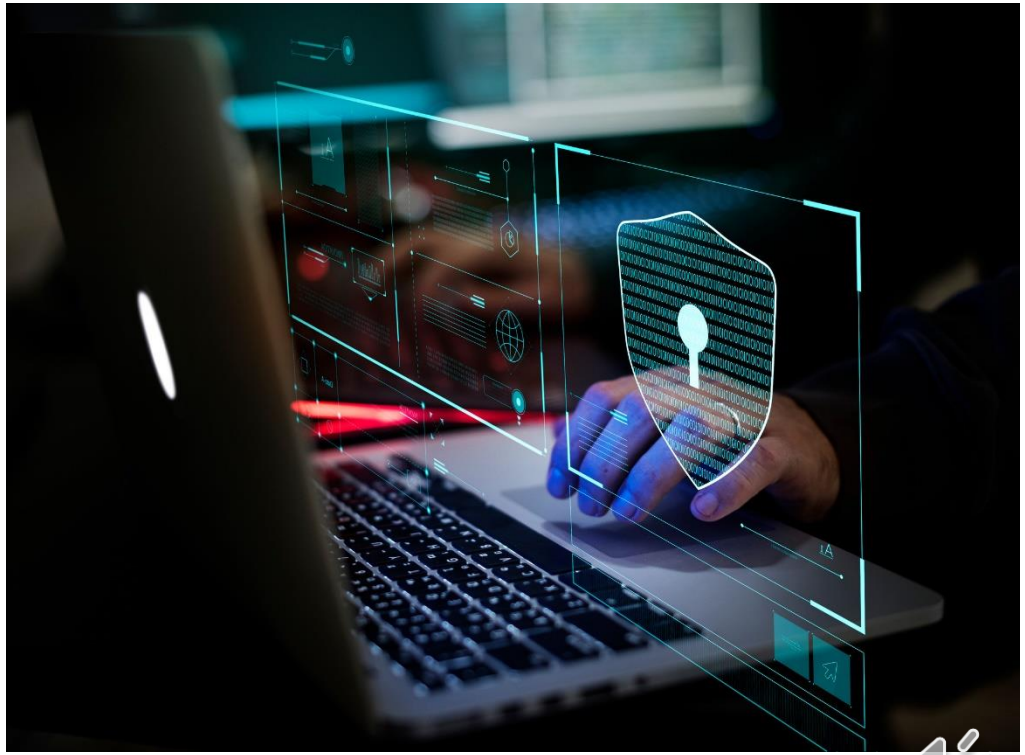
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Security Part4 – STM32 security in practice

Step by step creation of secure bootloader with software authentication

- Firmware
  - Confidentiality
  - Integrity
  - Authenticity
- Secure boot
  - Simple secure bootloader
  - Trusting the secure bootloader
  - Authenticate target application from bootloader
- Introduction to secure firmware update



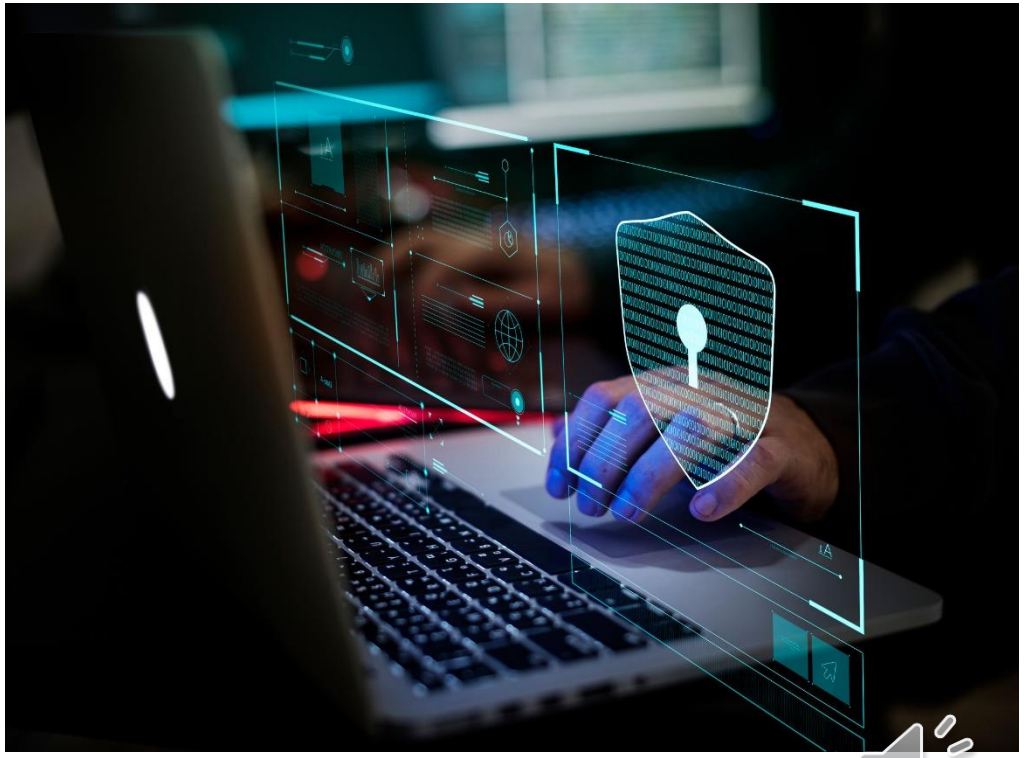
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Security Tips

Set of short tips related to STM32 security configuration and usage

- STM32 memory isolation purpose
- STM32L5 TrustZone and RDP safe usage
- ... to be continued



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Conclusion

Let’s start our adventure

Whole list of our on-line sessions you can find at:

- www.st.com/stm32education , section MOOC

You can find there all information about the sessions: agenda, list of prerequisites, links to download the materials

We would be happy to hear from you and answer your question at our community web page:

- https://community.st.com

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Thank you

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